Stelios Zacharioudakis

MACHINE LEARNING ENGINEER | AI AGENTS & REINFORCEMENT LEARNING

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Machine learning engineer combining production software ownership with AI agents, reinforcement learning and multilingual NLP. Led architecture and delivery of a deployed association platform; build learning systems, reproducible evaluations and reliable applications from model integration through operation.

PROFESSIONAL EXPERIENCE

Paphos Medical Association - Head Engineer (Pro Bono) | Jun 2022 - Jul 2026

- Led end-to-end delivery of AsklepiosMed, consolidating member services and association administration in a donated production platform.
- Designed React interfaces, Node.js/Express APIs and PostgreSQL persistence; owned role-based access, administrative workflows and Linux deployment.

SELECTED ENGINEERING PROJECTS

AsklepiosMed - Multilingual NLP workflows

- Integrated LLM-assisted drafting, email preparation and text revision into administrative publishing, turning short notes into structured content for editor review.
- Delivered Greek-to-English, Russian and Simplified Chinese translation workflows with terminology checks, output validation, fallback behavior and token-usage accounting.

ForgeRL / ForgeBench - Reinforcement learning and agent evaluation

- Built a coding-agent evaluation platform with an offline fitted-Q routing controller, isolated execution and durable budget controls; compared six policies in a 162-episode transfer pilot and a separate 1,296-attempt follow-up.

Inference Systems Lab - GPU serving evaluation

- Benchmarked 16,384 vLLM requests across RTX 4090/A100, two models and four repeats; measured 1.23-1.47x cached/uncached throughput ratios for shared-prefix workloads at concurrency 32, with output mismatches disclosed.

MTA-Scan - Streaming data and temporal evaluation

- Built a monitored GTFS-Realtime pipeline and FastAPI/Mapbox workspace with bounded storage and forecast-expiry controls; verified 44 daily exports and 28 evaluated days. Added a separate official service-alert revision archive and route-filtered context.

MRI reconstruction - PyTorch and generative models | BSc thesis: Excellent, 10/10

- Reimplemented a published score-based reconstruction method with Flax-to-PyTorch checkpoint conversion; reproduced reference PSNR within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling.
- Evaluated reconstruction quality, uncertainty and retrospective domain shift; published reproducible evidence and independent CPU ensemble diagnostics.

TECHNICAL SKILLS

ML & AI: Python, PyTorch, reinforcement learning, LLM agents, world models, generative models, NLP, vLLM, NumPy, SciPy, scikit-learn and experimental evaluation.

Engineering: FastAPI, React, Node.js/Express, JavaScript, PostgreSQL, Redis, Linux, Nginx, Docker, Git and automated testing.

EDUCATION

BSc in Computer Science | National and Kapodistrian University of Athens, Jun 2026

Degree: Very Good | Thesis: Excellent (10/10), deposited September 2026.

Independent technical work: verifier-guided self-improvement, continual learning for coding agents, latent world models and policy evaluation; ongoing and unpublished.